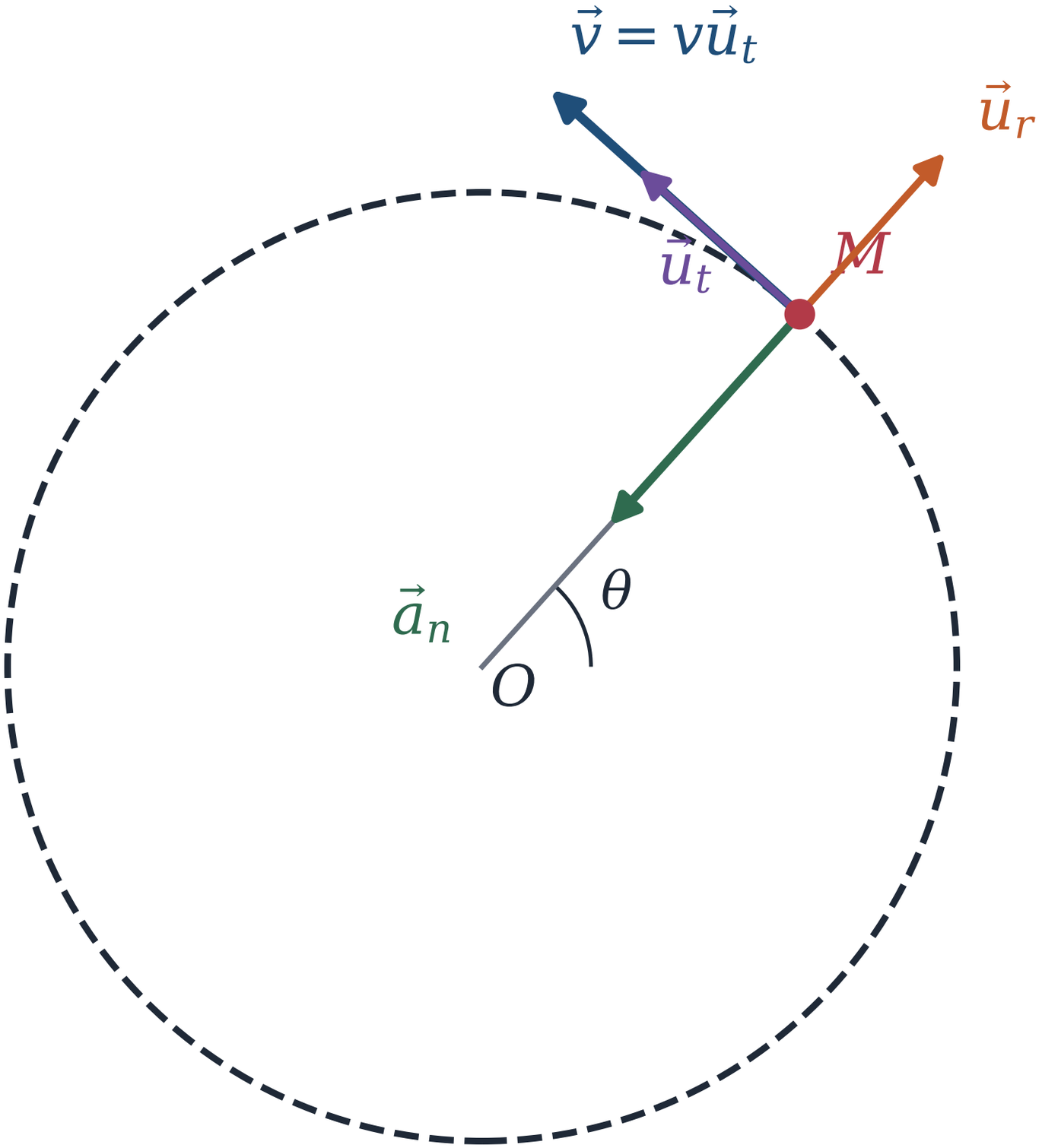

$$a_t = \dot{v}, \qquad a_n = \frac{v^2}{R}$$